

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the `Processed_KEY` folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the `Processed_KEY` folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(cowplot)
```

```
##  
## Attaching package: 'cowplot'  
##  
## The following object is masked from 'package:lubridate':  
##  
##     stamp
```

```
# verify home directory  
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
# read in NTL-LTER dataset  
peter_paul_chem_nutrients <- read.csv(here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul"))
```

```
# read in Niwot Ridge litter dataset  
litter_data <- read.csv(here("Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"))
```

```
#2  
class(peter_paul_chem_nutrients$sampledte) # character
```

```
## [1] "character"
```

```
class(litter_data$collectDate) # character
```

```
## [1] "character"
```

```
peter_paul_chem_nutrients$sampledte <- ymd(peter_paul_chem_nutrients$sampledte)  
litter_data$collectDate <- ymd(litter_data$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```

#3
my_theme <-
  theme(
    # plot background
    plot.background = element_rect(fill = "white"),
    panel.background = element_rect(fill = "white"),

    # set fonts
    plot.title = element_text(size = 14),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "black"),
    legend.text = element_text(size = 10, color = "black"),

    # axis settings and gridline
    axis.ticks = element_line(linewidth = 0.25),
    axis.line = element_line(color = "black", linewidth = 0.5),
    panel.grid = element_blank(),

    # legend
    legend.position = "bottom",

    # adjust margins for display
    plot.margin = margin(10, 20, 20, 10)
  )

```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```

#4
plot_q4 <- ggplot(data = peter_paul_chem_nutrients,
  mapping = aes(x = po4, y = tp_ug, color = lakename)) +
  geom_point(size = 0.75) +
  geom_smooth(method = "lm",
    se = FALSE,
    linewidth = 0.75) +
  scale_x_continuous(limits = c(0, 50),
    expand = c(0, 0)) +
  scale_y_continuous(limits = c(-10, 150),
    expand = c(0, 0)) +
  scale_color_manual(values = c("#41b6c4", "#0c2c84")) +
  my_theme +
  labs(
    x = "Phosphate",
    y = "Total Phosphorus",
    color = element_blank(),
    title = "Phosphorous vs. Phosphate Levels in Peter and Paul Lakes\n"
  )

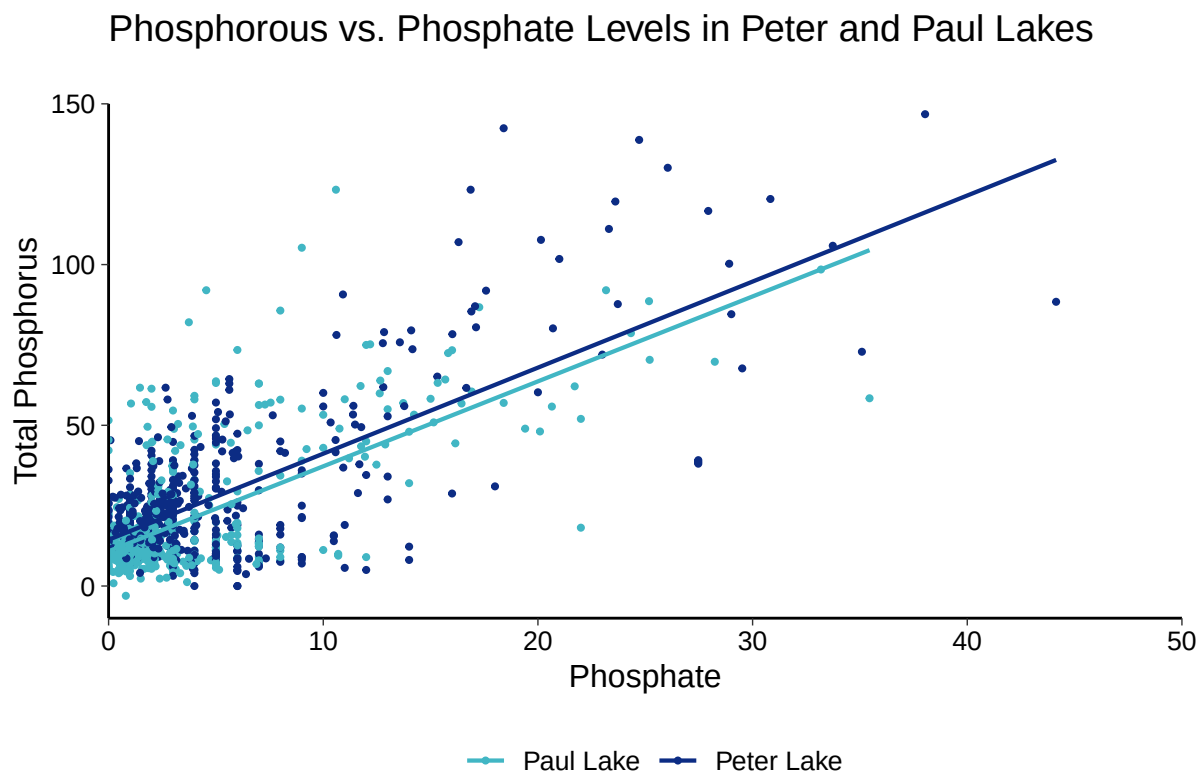
```

```
plot_q4
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 21947 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

```
## Warning: Removed 21947 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
```

```
# make base plot  
# add labels to month as a factor for visualization purposes  
base_boxplot <- peter_paul_chem_nutrients %>%
```

```

mutate(month = factor(month, levels = c(1:12), labels = month.abb)) %>%
filter(month != "Feb") %>% # exclude February b/c there's no data this month
ggplot(aes(x = month, color = lakename)) +
my_theme +
scale_color_manual(values = c("#41b6c4", "#0c2c84")) +
labs(x = element_blank()) +
theme(legend.position = "None",
      axis.title.y = element_text(size = 8), # make smaller just for this plot
      plot.margin = margin(0, 10, 0, 10)) # adjust margins just for this plot

# make each of the three boxplots
# Temperature
temp_plot <- base_boxplot +
  geom_boxplot(aes(y = temperature_C)) +
  labs(y = "Temperature (C)\n")

# Phosphorous
tp_plot <- base_boxplot +
  geom_boxplot(aes(y = tp_ug)) +
  labs(y = "Total Phosphorous\n")

# Nitrogen
tn_plot <- base_boxplot +
  geom_boxplot(aes(y = tn_ug)) +
  labs(y = "Total Nitrogen\n",
       color = element_blank()) +
  theme(legend.position = "bottom")

# combine plots using cowplot
plot_q5 <- plot_grid(temp_plot, tp_plot, tn_plot,
                    ncol = 1,
                    align = "h")

```

```

## Warning: Removed 3550 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

## Warning: Removed 20713 rows containing non-finite outside the scale range
## ('stat_boxplot()').

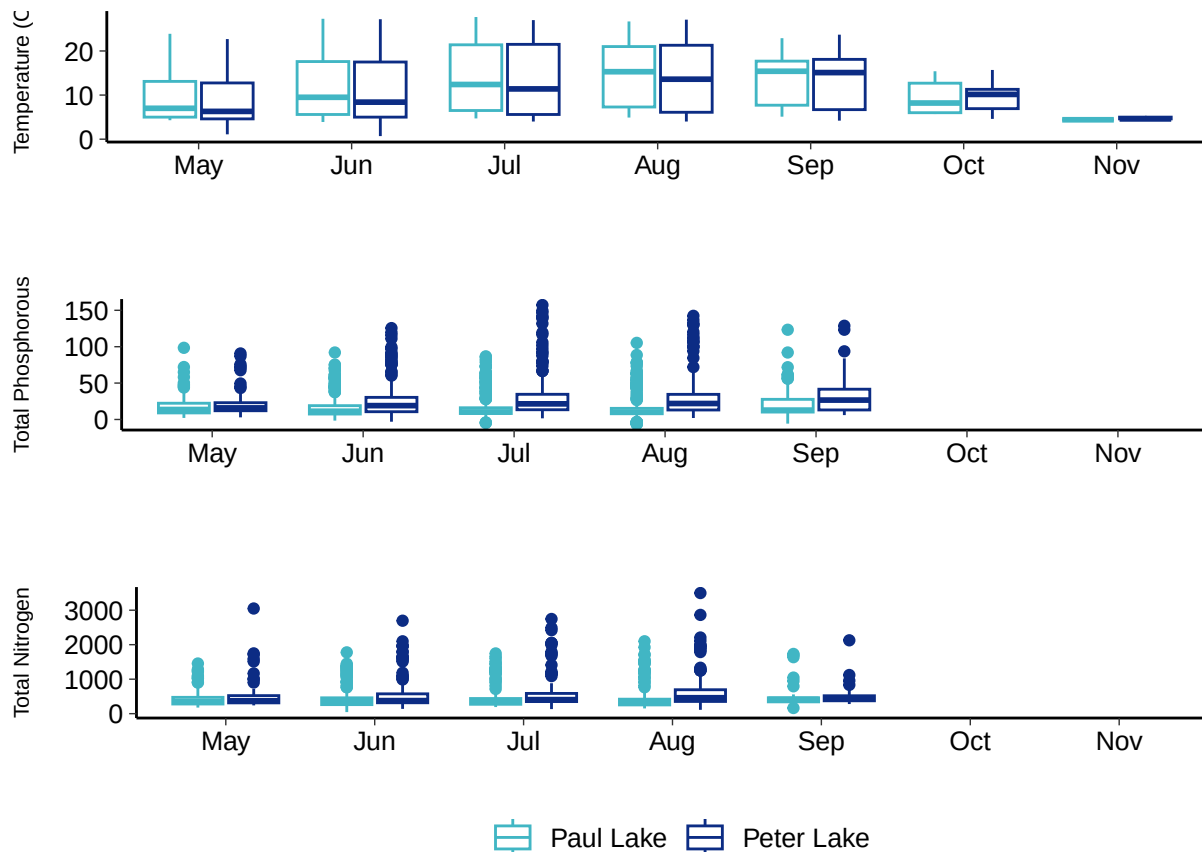
```

```

## Warning: Removed 21567 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```
plot_q5
```



Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: For temperature, we observe a clear trend where median temperature in both lakes increases in the summer and decreases in the spring/fall. For most months, temperature trends slightly lower in Peter Lake than Paul Lake. There is not a clear seasonal trend for phosphorous and nitrogen levels, but we do observe overall higher levels of both phosphorous and nitrogen in Peter Lake. We also see more outlier observations of these metrics than of temperature.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

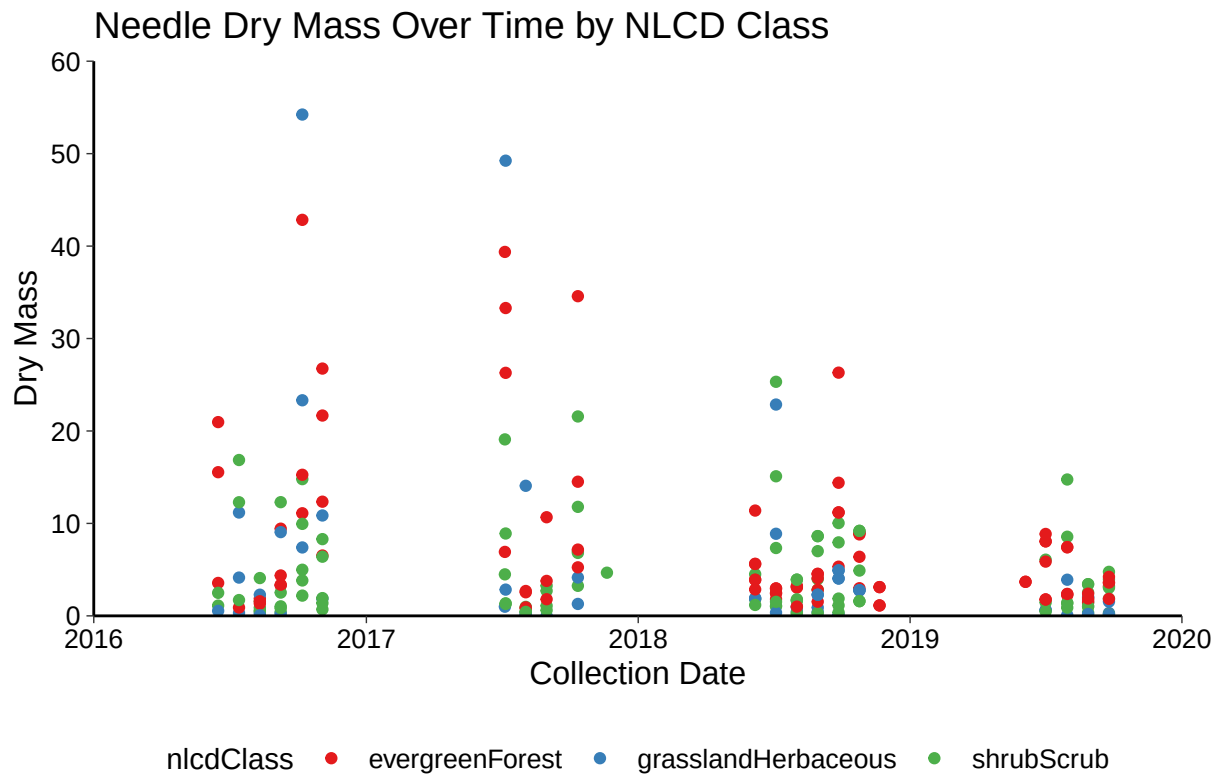
```
# set up base plot for questions 6 and 7
base_plot_needles <- litter_data %>%
  filter(functionalGroup == "Needles") %>%
  ggplot(aes(x = collectDate, y = dryMass)) +
  scale_x_date(limits = ymd(c("2016-01-01", "2020-01-01")),
               date_breaks = "1 year",
               date_labels = "%Y",
               expand = c(0, 0)) +
  scale_y_continuous(limits = c(0, 60), expand = c(0, 0)) +
  my_theme +
  labs(x = "Collection Date",
       y = "Dry Mass",
```

```
title = "Needle Dry Mass Over Time by NLCD Class")
```

```
#6
```

```
plot_q6 <- base_plot_needles +  
  geom_point(aes(color = nlcdClass)) +  
  scale_color_brewer(palette = "Set1")
```

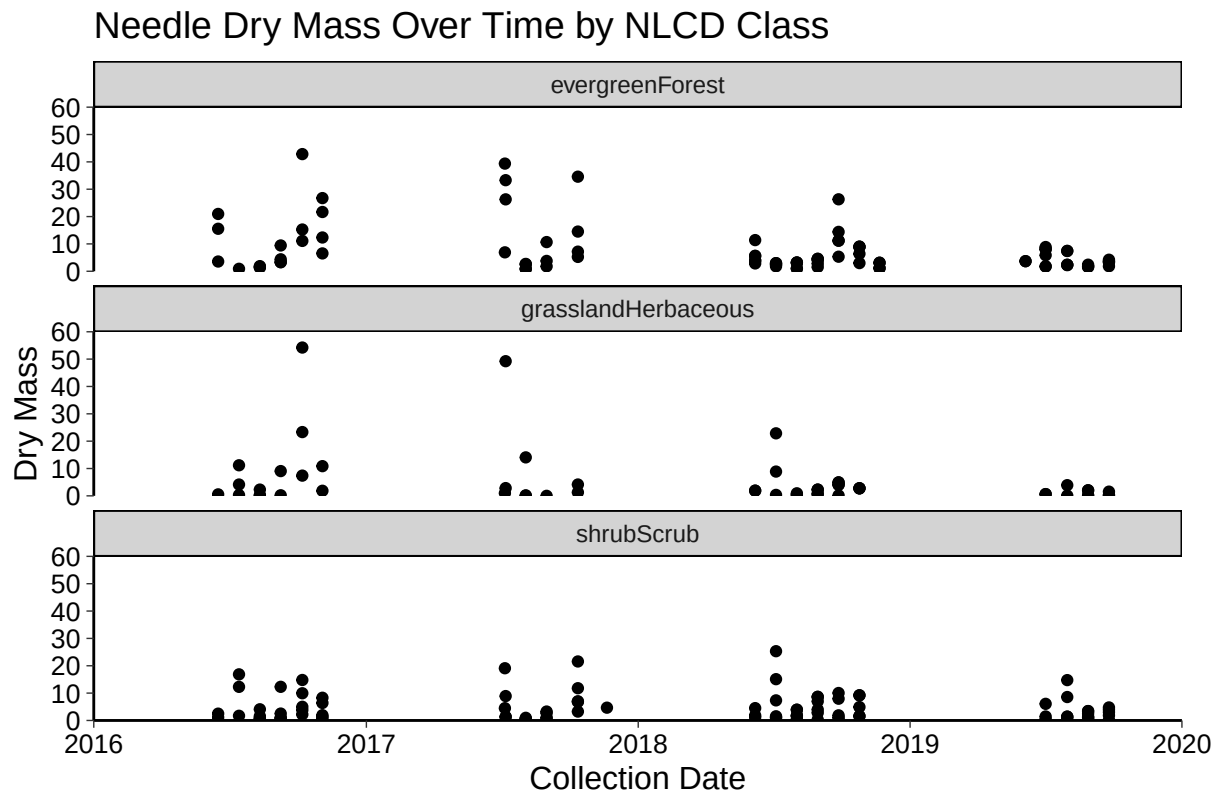
```
plot_q6
```



```
#7
```

```
plot_q7 <- base_plot_needles +  
  geom_point() +  
  facet_wrap(facets = "nlcdClass", ncol = 1) +  
  theme(strip.background = element_rect(fill = "lightgrey", color = "black"))
```

```
plot_q7
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: Plot 7 (separated by facets) is the most effective visualization. Faceting by NLCD class allows you to more easily see how needle dry mass varies over time in each of the land types while still allowing you to do a side-by-side comparison between NLCD classes. Separating land class by color is not effective as the plot is very busy and it's difficult to see any trend over time within a single land class.